Ruel & Nima

Does reaction time change throughout the day?

Reaction time may be classified as a measure of general mental cognition, leading to broader inferences.

State:

We want to determine whether there is a decrease in reaction speed from 1st period to 4th period.

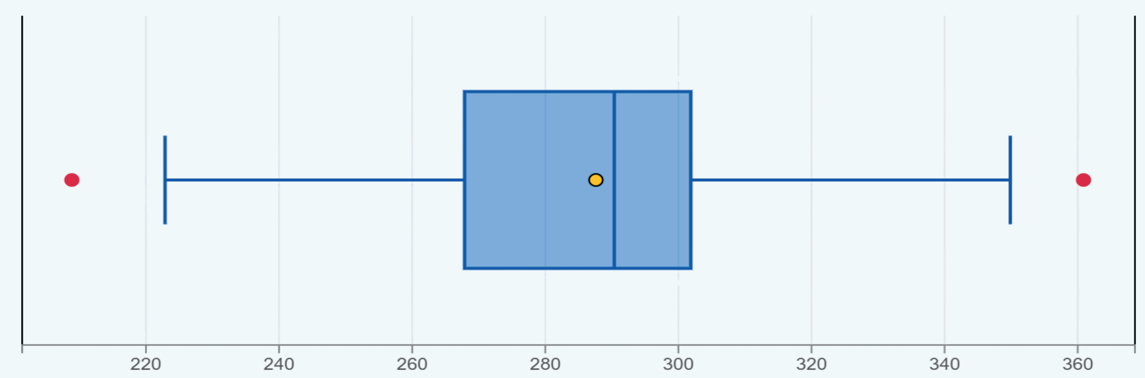
μ = mean difference in reaction time between the morning and the afternoon (ms)

(afternoon's measurement − morning's measurement)

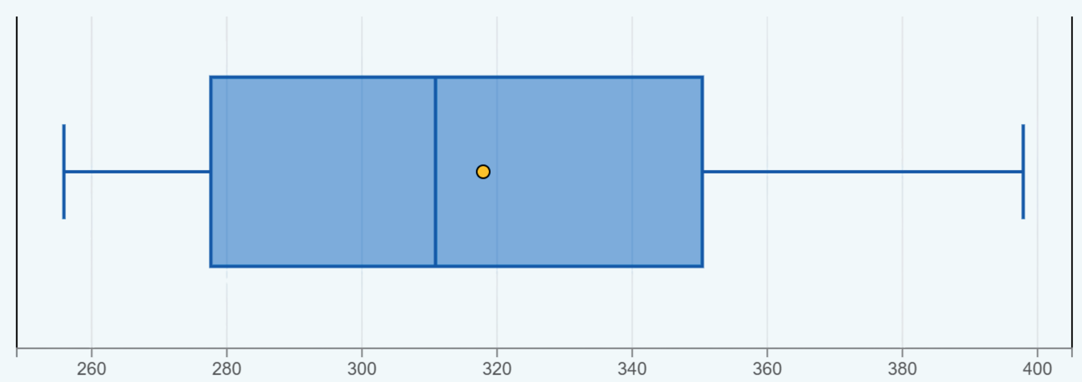
H_0: μ = 0 ms

H_a: μ > 0 ms

Morning reaction times



Afternoon reaction times



Plan:

We want to perform a one-sample t-test for the mean difference in reaction times between the morning and the afternoon with a significance level of 0.05.

"SRS"

Random ✓

10 · n ≤ N

10 · 30 ≤ 1500+

300 ≤ 1500+ ✓

Independence ✓

n ≥ 30

30 ≥ 30 ✓

Normality ✓

A stratified random sample of clusters was taken, selecting one class from each floor of the school, along with the 600 wing. A SRS of 6 students was then taken from each class, for a total sample size of 30 students.

Each student took a reaction speed test during 1st period and 4th period. Their reaction times, in milliseconds, was recorded as a pair.

Do:

$$\sigma = \frac{s}{\sqrt{n}} = \frac{46.42}{\sqrt{30}} = 8.48\ ms$$

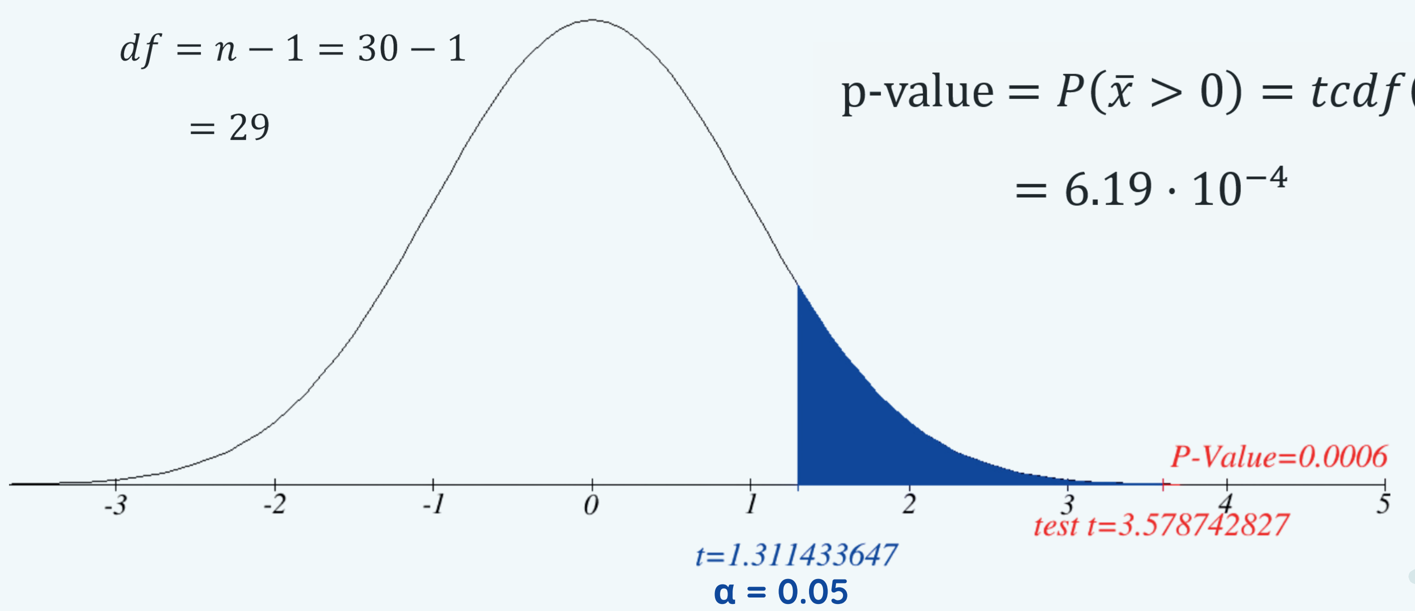
$$t = \frac{\bar{x} - \mu_0}{\sigma} = \frac{30.\overline{3} - 0}{8.48} = 3.58$$

$df = n - 1 = 30 - 1$

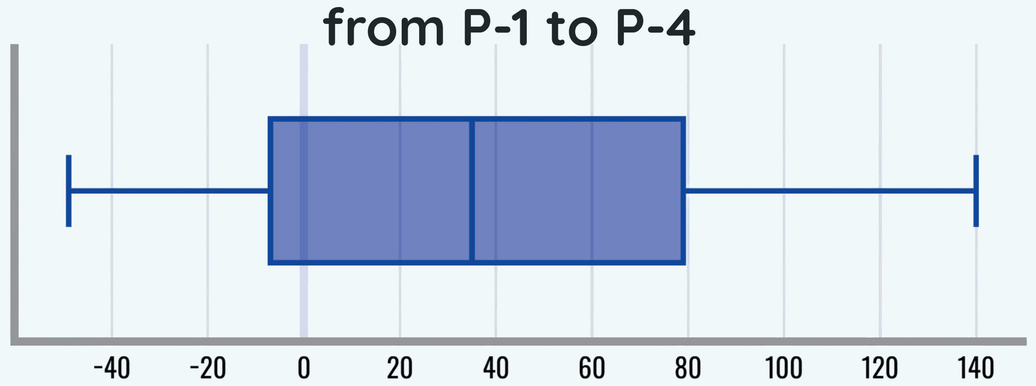
$= 29$

$$\text{p-value} = P(\bar{x} > 0) = tcdf(lower = 3.58, upper = \infty, df = 29)$$

$$= 6.19 \cdot 10^{-4}$$



Difference in reaction times from P-1 to P-4



Reaction times of each student

Region	P1 Time (ms)	P4 Time (ms)	Difference (ms)
1st Floor	287	379	92
	233	320	87
	209	349	140
	327	331	4
	361	351	-10
	302	381	79
2nd Floor	267	268	1
	223	268	45
	245	280	35
	276	356	80
	271	312	41
	302	310	8
3rd Floor	293	269	-24
	350	301	-49
	245	276	31
	274	277	3
	287	275	-12
	300	303	3
4th Floor	301	341	40
	291	370	79
	290	387	97
	347	333	-14
	327	378	51
	315	398	83
600 Wing	293	287	-6
	316	281	-35
	302	343	41
	290	256	-34
	255	274	19
	253	288	35

Conclude:

If there were truly no difference in reaction time between 1st and 4th period, the probability of obtaining a sample mean greater than or equal to $30.\overline{3}$ is 0.00062.

Since the p-value of 0.00062 is less than the significance level of 0.05, we reject the null hypothesis. It is quite unlikely that there is no difference between reaction speed in the morning and the afternoon.

Since students were randomly sampled from the population, our results can be generalized to all VanTech students. Cause and effect can not be established because experimental treatments were not assigned.

Possible sampling biases may be underrepresentation of students who aren't present in class and biases between age and gender due to clustered sampling. Between 1st and 4th period, variables confounding with the time of day may be the difference in class rigor, sleep quality, number of free blocks, and other factors affecting mental cognition